

DocsGuide Literature Review

Introduction

DocsGuide is an AI-powered chatbot designed to assist Nepali citizens in navigating complex government document procedures such as obtaining citizenship certificates, passports, and birth registrations. It provides clear, step-by-step guidance, lists the required documents, and offers conversational support to clarify legal questions and concerns. By combining natural language understanding with verified government data, DocsGuide aims to reduce confusion, minimize delays, and make essential services more accessible to the public.

High-Level Problem (Research Question)

The core research question is: *How can a RAG based AI-driven conversational system deliver accurate and user-friendly assistance to simplify the process of obtaining official government documents for Nepali citizens?*

General Review

Government document procedures in Nepal—such as obtaining citizenship certificates, passports, and birth registrations—are often complex due to inconsistent information, language diversity, and evolving legal requirements. Building an AI-powered conversational assistant to simplify these processes requires understanding key concepts like *retrieval-augmented generation (RAG)*, *natural language understanding (NLU)*, *text embeddings*, *vector databases*, and *multilingual large language models (LLMs)*.

Key Elements

- **Major Ideas:** Retrieval-Augmented Generation (RAG), multilingual embeddings, LLaMA-based Nepali fine-tuning, conversational AI for government services, knowledge base creation, and dynamic data updates.
- **Available Datasets:** No large open-source Nepali government-procedure datasets exist; data must be manually collected from official portals, gazettes, and legal documents.
- **Common Models Used:** Meta LLaMA (various sizes), mT5, XLM-R for multilingual understanding, and Sentence Transformers for embeddings. For retrieval, FAISS, ChromaDB and Pinecone are standard choices.

- **Major Issues:**
 - **Language:** Nepali morphology and dialect variations can lower embedding and LLM accuracy.
 - **Model Accuracy:** Even fine-tuned LLaMA models may misinterpret domain-specific legal terms.
 - **Dataset Acquisition:** Official data is often scattered, outdated, or inaccessible; frequent updates are required.
 - **Legal/Ethical Constraints:** Restrictions on automated scraping and data-sharing must be strictly followed.

Paper Review

Paper 1 : NepKanun: A RAG-Based Nepali Legal Assistant (“NepKanun: A RAG-Based Nepali Legal Assistant”)

- **Authors:** Not Mentioned
- **DOI/Link:** <https://openreview.net/pdf?id=LuXTBI6GSh>
- **Main Idea & Contribution:** Addresses the lack of accessible legal information in Nepali by building a RAG-based QA system. A novel system combining a fine-tuned LLM with RAG, and a curated dataset of 10k Nepali legal QA pairs.
- **Dataset:** 10k high-quality Nepali legal QA pairs. Scraping (Supreme Court website, news), OCR of legal PDFs, manual validation.
- **Model:** Fine-tuned LLaMA 3.2 3B. Embeddings: multilingual SentenceTransformer. Vector DB: ChromaDB.
- **Procedure/Configuration:** Data curation, LoRA/QLoRA fine-tuning of LLaMA , RAG setup with ChromaDB for retrieval, Generation.
- **Techniques/Tricks:** Parameter-Efficient Fine-Tuning (PEFT) via LoRA/QLoRA, OCR for digitizing scanned documents, MMR for diverse retrieval.
- **Performance Metrics:** BERTScore F1: 0.82 (simple), 0.77 (moderate), 0.71 (complex). Human evaluation showed high scores for Faithfulness (4.5/5) and Relevance (5/5).
- **Key Learning:** RAG is exceptionally effective for low-resource, high-stakes domains like law, as it grounds answers in source texts, reducing hallucinations. Performance degrades with query complexity, indicating a limit in reasoning ability.

Paper 2: LawPal: A RAG-Based System for Legal Accessibility in India (Panchal et al.)

- **Authors:** Dnyanesh Panchal, Aaryan Gole, Vaibhav Narute, Raunak Joshi

- **DOI/Link:** <https://arxiv.org/abs/2502.16573v1>
- **Main Idea & Contribution:** The paper's core aim is to enhance legal accessibility in India by developing a Retrieval-Augmented Generation (RAG) based chatbot. Its key contribution is demonstrating a scalable and efficient system that integrates FAISS for retrieval and DeepSeek models for both embeddings and generation, ensuring high accuracy and speed.
- **Dataset:** Legal books, Indian Constitution, case archives, government sites (structured/unstructured text).
- **Model:** DeepSeek embeddings and DeepSeek-R 1.5B model for generation.
- **Procedure/Configuration:** Data scraping, Preprocessing (OCR, chunking, NER), Embedding with DeepSeek, FAISS indexing, RAG generation with DeepSeek-R.
- **Techniques/Tricks:** Use of FAISS for highly scalable similarity search, processing a mix of structured/unstructured legal texts.
- **Performance Metrics:** >90% legal accuracy, strong BLEU/ROUGE/LCS scores. The system demonstrated efficient retrieval times of 10-50ms with FAISS and response generation times ranging from 800-1500ms with DeepSeek-R 1.5B, emphasizing scalability and efficient retrieval.
- **Key Learning:** The choice of vector database (like FAISS) is critical for building scalable and responsive production-level legal AI systems that can handle large volumes of data. This paper highlights how RAG, by augmenting a language model with external knowledge, can overcome the limitations of a static model, reducing factual inaccuracies and "hallucinations" by grounding the responses in up-to-date and specific legal documents.

Paper 3: Bidirectional English-Nepali Machine Translation System for the Legal Domain (Poudel et al.)

- **Authors:** Shabdapurush Poudel, Bal Krishna Bal, Praveen Acharya
- **DOI/Link:** <https://aclanthology.org/2024.sigul-1.7/>
- **Main Idea & Contribution:** This paper addresses the significant lack of digital resources for the Nepali language, particularly within the legal domain. The authors' main contribution is the creation of a domain-specific parallel corpus of 125,000 sentences. They then use this corpus to train a Neural Machine Translation (NMT) system to enable bidirectional translation between English and Nepali legal texts.
- **Dataset:** A custom-built legal corpus consisting of 125,000 parallel sentences.
- **Model:** A Neural Machine Translation (NMT) system with an encoder-decoder architecture, utilizing the fairseq toolkit.

- **Procedure/Configuration:** The methodology involved a two-step process: first, the collection and development of a new legal corpus from various sources, and second, the training of an NMT model on this dataset. This approach demonstrates that a custom-curated dataset can effectively overcome the challenges of a low-resource language domain.
- **Techniques/Tricks:** The core technique is the creation of a domain-specific dataset. This is a crucial trick for enabling effective NMT in specialized fields with limited existing resources.
- **Performance Metrics:** The system achieved a BLEU score of 7.98 for Nepali to English translation and 6.63 for English to Nepali.
- **Key Learning:** Despite the challenges of a low-resource language like Nepali, a focused effort to build a high-quality, domain-specific dataset is a viable and effective strategy for creating a functional machine translation system. The performance metrics also provide a valuable baseline for what can be achieved with this approach.

Paper 4: Pre-trained Language Models for the Legal Domain: A Case Study on Indian Law (Paul et al.)

- **Authors:** Shounak Paul, Pawan Goyal
- **DOI/Link:** <https://arxiv.org/pdf/2209.06049>
- **Main Idea & Contribution:** This paper investigates the effectiveness of adapting existing pre-trained language models (PLMs) to a specific legal domain, in this case, Indian law. The main contribution is demonstrating that continued pre-training of popular legal PLMs like LegalBERT and CaseLawBERT on Indian legal text significantly enhances performance on Indian legal NLP tasks. The study also shows that this approach can generalize and even improve performance on tasks from other legal domains (EU, UK).
- **Dataset:** A custom corpus of Indian legal data.
- **Model:** LegalBERT and CaseLawBERT (pre-trained language models) were used for continued pre-training. The authors also trained a model from scratch.
- **Procedure/Configuration:** The methodology involved three primary experiments: continued pre-training of LegalBERT on Indian data, continued pre-training of CaseLawBERT on Indian data, and training a new model from scratch with a vocabulary based on Indian legal text. The performance of these models was evaluated on three benchmark legal NLP tasks: Legal Statute Identification, Semantic Segmentation of Court Judgments, and Court Appeal Judgment Prediction.
- **Techniques/Tricks:** The core technique is "continued pre-training," also known as domain adaptation. This approach leverages a base model's existing knowledge and fine-tunes it on a new, specific dataset, which is more efficient

than training a model from scratch. The authors also highlight the importance of a domain-specific vocabulary.

- **Performance Metrics:** The paper reports enhanced performance on both Indian and non-Indian datasets. Specific scores are not provided in the abstract, but the overall claim is a significant improvement over baseline models.
- **Key Learning:** The paper teaches us that when working with a new domain that has similar characteristics to an existing one, adapting a pre-trained model through continued pre-training is a very effective strategy. It offers a powerful alternative to building a model from scratch, saving significant time and computational resources while still achieving strong results and even improving performance on the original domain.

Paper 5 : Domain-adaptative Continual Learning for Low-resource Tasks: Evaluation on Nepali (Duwal et al.)

- **Authors:** Sharad Duwal, Suraj Prasai, Suresh Manandhar
- **DOI/Link:** <https://arxiv.org/pdf/2412.13860>
- **Main Idea & Contribution:** This paper evaluates the effectiveness of Domain-Adaptive Pre-training (DAPT) in a low-resource setting, using the Nepali language as a case study. The key contribution is the demonstration that by continually training a pre-trained large language model (LLM) on synthetic data, you can significantly enhance its performance for a new, low-resource language, even while observing some "forgetting" of its original knowledge.
- **Dataset:** The model was trained using a corpus of synthetic Nepali data. The paper emphasizes the use of this synthetically generated data to overcome the inherent scarcity of real-world Nepali text, which is a key aspect of their approach.
- **Model:** Llama 3 8B, a pre-trained large language model.
- **Codebase:** <https://huggingface.co/Wiseyak/OpenWiseyak-0.1-Base-4bit>
- **Procedure/Configuration:** The authors applied a continual learning approach to a Llama 3 8B model using 4-bit QLoRA for efficient fine-tuning. They then evaluated the adapted model on various benchmarks to measure performance, knowledge acquisition, and the extent of forgetting compared to the base model.
- **Techniques/Tricks:** The core techniques are Domain-Adaptive Pre-training (DAPT), which is a form of continual learning, and the use of QLoRA for memory-efficient training. The use of synthetically generated data is a key trick for overcoming the scarcity of real-world data in a low-resource language.
- **Performance Metrics:** The adapted model showed a performance increase as high as 19.29% compared to the base model's 4.98% increase when evaluated with an increasing number of shots. This suggests that the model has a latent retention of the new knowledge.
- **Key Learning:** The paper shows that for low-resource languages, adapting a powerful existing model is a viable and effective approach, especially when coupled with

memory-efficient techniques and synthetic data. This is a powerful alternative to collecting massive real-world datasets or training a model from scratch.

Literature Review Matrix

Title/Author/ Date	Conceptual Framework	Research Question(s)/ Hypotheses	Datasets	Methodology	Analysis & Results	Conclusions	Implications for Future Research
NepKanun: A RAG-Based Nepali Legal Assistant	RAG pipeline with fine-tuned LLaMA for Nepali legal QA.	Can a domain-specific RAG system improve access to Nepali legal info?	• 10k Nepali legal QA pairs (SC website, laws, news)	• Data scraping & OCR • Manual validation & cleaning • LoRA fine-tuned LLaMA 3.2B • RAG pipeline with ChromaDB embeddings	• BERTScore F1: 0.82 (simple), 0.71 (complex) • Human eval: high faithfulness/relevance	Works well for simple queries, weaker on complex reasoning.	• Expand dataset • Improve reasoning • Cover more legal areas
LawPal: A RAG-Based System for Legal Accessibility in India (Panchal et al., 2025)	RAG chatbot integrating FAISS + DeepSeek embeddings for Indian legal texts.	Can RAG+FAISS provide accurate, fast, and accessible legal info in India?	• Indian Constitution, case archives, law books, govt. sites	• Data scraping & API collection • OCR, chunking & NER preprocessing • Embeddings with DeepSeek-R1:5B • FAISS-based vector retrieval • RAG response generation	• >90% legal accuracy • Strong BLEU/ROUGE/LCS • Efficient retrieval at scale	Increases legal literacy, reduces misinformation.	• Jurisdiction-specific modules • Multilingual expansion • Bias mitigation

Bidirectional English–Nepali MT for Legal Domain (Poudel, Bal & Acharya, 2024)	Transformer-based NMT with encoder-decoder for legal text.	Can a bidirectional NMT system handle English↔Nepali legal translation effectively?	• 125k English–Nepali parallel corpus (manual, SC, news)	• Manual translations & web scraping • Data preprocessing (IndicNLP, Sacremoses, BPE) • RNN baseline tested • Transformer-based NMT via Fairseq	• BLEU: 7.98 (Nep→Eng), 6.63 (Eng→Nep) • Better than RNN baseline	First robust legal MT for Nepali; improved fluency.	• Expand corpus • Domain adaptation • Hybrid MT/LLM approaches
Pre-trained LMs for Legal Domain: Indian Law Case Study (Paul et al., IIT Kharagpur, 2023)	Domain-specific PLMs (LegalBERT, CaseLawBERT) retrained on Indian law.	Does continued pretraining on Indian law improve PLM performance across tasks?	• 5.4M docs (27GB) – SC & HC judgments, statutes (1950–2019)	• Collect SC/HC cases & statutes (~27GB) • Preprocess (cleaning, chunking, tokenization) • Retrain LegalBERT & CaseLawBERT • Train CustomInLawBERT from scratch • MLM & NSP training objectives	• InLegalBERT best on 4/5 tasks (statute ID, segmentation, judgment prediction)	Domain-specific pretraining boosts legal NLP performance.	• Larger-scale pretraining • More downstream tasks • Multilingual legal LMs
Domain-adaptive Continual Learning for Low-resource	Domain-adaptive continual pretraining (DAPT) on	Can continual learning with synthetic	• 5M Nepali–English pairs • 114K	• Generate synthetic parallel & instruction data • chrF++ filtering	• Improved Nepali generation (grammar, correctness)	DAPT with synthetic data enables	• Extend to other South Asian low-resource

e Tasks: Evaluation on Nepali (Duwal, Prasai & Manandhar – Wiseyak, 2024)	Llama 3 8B using synthetic Nepali-English data in QLoRA.	data effectively adapt LLMs to Nepali without major loss of prior knowledge?	translated instruction triplets	for quality • QLoRA continual pretraining (translation + bilingual token prediction) • Supervised finetuning on instructions	• Some English forgetting, but better few-shot retention (+19.29% ARC-Challenge) • Learned Nepali syntax (adjective–noun dependencies)	Nepali adaptation with reasonable retention.	new languages • Add human evaluation & Nepali benchmarks
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Project Solution Proposal

1. Data Collection and Curation

Acquisition:

- Manually collect government documents and official guidelines for key procedures:
 - **Citizenship Certificate:** Citizenship Act 2063 (Nepali & English), DAO guidelines (Bhaktapur, Darchula), Nepal Law Commission references.
 - **Birth Registration Certificate:** Births, Deaths, and Other Personal Events (Registration) Rules, municipal office guides (e.g., Gokarneshwor).
 - **Passport:** Passport Act 2019, Passport Regulation 2020, Department of Passport guidelines.
- Sources include PDFs, official websites, publications, and scanned forms.

Verification:

- Cross-check content across multiple sources for accuracy and consistency.
- Assign verifiers to validate passages; store provenance metadata (source, date, page, verifier, jurisdiction).

Structuring:

- Clean text, normalize legal terms, and remove unnecessary headers/footers.
- Tag each chunk with metadata: procedure type, language, document type, source ID, jurisdiction, verification status.

2. Embedding and Indexing

Embedding:

- Use **Sentence Transformer: paraphrase-multilingual-mpnet-base-v2** to convert passages into vector embeddings.

Indexing:

- Store embeddings and meta data in a **ChromaDB vector database** for high-speed similarity search.
- Similarity metric = cosine similarity.

3. RAG Pipeline

User Query:

- Accept natural language queries in Nepali or English; auto-detect language.

Retrieval:

- Embed the query and search in vector database for the most relevant passages.
- Use passages from verified government documents listed above.

Generation:

- Pass retrieved passages and user query to **OpenWiseYak (LLaMA-3 8B, LoRA-fine-tuned for Nepali language)**.
- Model generates:
 - Step-by-step procedural guidance.
 - List of required documents.
 - Appropriate office/authority to visit.
- Always cite sources from retrieved passages.

4. Deployment and Maintenance

Interface:

- Bilingual web chatbot interface with simple UX.
- Display step-by-step instructions, required documents, office details, and citations.

Updates:

- Periodically update the knowledge base with new government documents.
- Re-run embedding and re-index passages in ChromaDB.

Monitoring:

- Track:
 - Query latency (<10s target)
 - Answer accuracy (%)
 - Completeness of procedural steps (%)
 - Clarity of language (user-rated)
 - Bilingual success (%)
 - User satisfaction (1–5 scale)

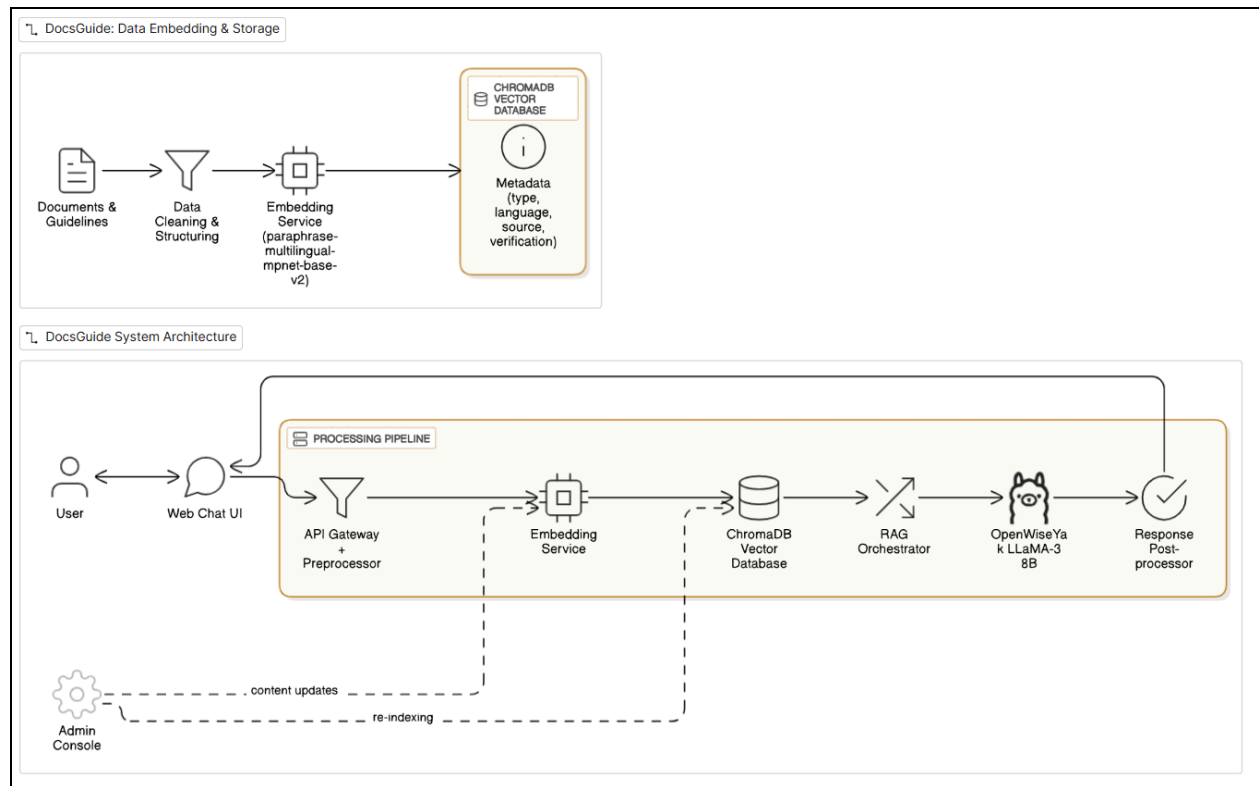


Figure: System Architecture Diagram

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